Answers and talk notes

Write and Solve Equations Using Multiplication or Division

What your student is learning

Tonight we solve equations where the variable was DIVIDED — by multiplying both sides by the same number.

Answers

Ask your student to explain before you look.

1. **C.** Multiplying by 6. Multiplying by 6 undoes dividing by 6.
2. **D.** 16. n = 16, because 8 × 2 = 16.
3. **B.** 20. x = 20, because 4 × 5 = 20.
4. **A.** 21. m = 21, because 7 × 3 = 21.
5. **B.** x ÷ 4 = 9. x ÷ 4 = 9 is solved by multiplying both sides by 4.

Talk about it

Why do we multiply to undo dividing? What would go wrong if you divided by 4 again?

Home mission

Put the same number of beans in each of 6 cups, without counting out loud. Tell your student only how many are in one cup, such as 4. Can they say how many beans there are in all? Pour the cups together to check.

When they are stuck, ask

- Is the variable being multiplied or divided here?
- What is the opposite of that?
- What does your answer mean in the story?

Try not to say

- Multiply both sides by 6 to solve 6x = 42.
- Do the same operation you see in the equation.
- The answer is just a number, the units do not matter.